Chapter 9 • Similar Figures

# Solved Exercise 9.4

**1. Answer the four polygon-angle questions.**
(i) Sum of interior angles of a decagon: $(10-2)180°=8\times180°=1440°$.
(ii) Each interior angle of a regular hexagon: $(6-2)\frac{180°}{6}=120°$.
(iii) Each exterior angle of a regular pentagon: $\frac{360°}{5}=72°$.
(iv) If the interior-angle sum is 1260°, then $(n-2)180°=1260° \Rightarrow n-2=7 \Rightarrow n=9$ sides.

**2. Parallelogram sides AB=10 cm, AD=6 cm enclose 45°. Find area.**
$A = ab\sin\theta$
$= 10\times6\times\sin45°$
$= 30\sqrt{2}\approx42.43$ cm².

**3. In parallelogram ABCD, ∠DAB=70°. Find all other angles.**
∠ABC=110°, ∠BCD=70°, ∠CDA=110°: opposite angles are equal and adjacent angles supplementary.

**4. Explain the tessellation made by cutting a square diagonally and attaching right triangles; discuss its angles.**
The square halves are 45°–45°–90° triangles. Congruent right triangles can be rotated/reflected in pairs to form rectangles, and rectangles tessellate without gaps.
At any tessellation vertex the angles must total 360°. Four right angles give $4\times90°=360°$, and eight 45° angles give $8\times45°=360°$.
A single numerical "interior angle of the new shape" is not determined unless the attachment diagram and which vertex is intended are supplied.
**Tessellation condition=360° around each vertex; the question does not uniquely specify one new interior angle.**

**5. A 3–4–5 right triangle is repeatedly reflected to cover a square of area 3600 square units.**
Triangle area=$(\frac{1}{2})(3)(4)=6$ square units. Required triangle copies=$\frac{3600}{6}=600$.
Thus the intended area answer is 600 triangular copies . Area alone does not determine a minimum number of reflection operations; if exactly one reflection produced each additional copy after the first, that convention would require 599 reflections.
**600 triangles by area; the printed wording confuses tile count with reflection count.**

**6. Twenty-five regular hexagons have side 5 cm. Find total area and discuss outer perimeter.**
One area=$(\frac{3\sqrt{3}}{2})(5^2)=37.5\sqrt{3}$ cm². Total=$25\times37.5\sqrt{3}=937.5\sqrt{3}\approx1623.80$ cm².
If the intended arrangement is the 5×5 hex-grid block behind the printed perimeter answer, it has 56 shared sides: exposed edges=$25\times6-2\times56=38$, so perimeter=$38\times5=190$ cm.
If "perfect hexagon" instead means a regular hexagon similar in outline with 25 times the area, its side scale is 5 and perimeter is $6\times25=150$ cm. The arrangement must be specified to choose between these models.

**7. How many 1 m×1 m tiles cover a 12 m×15 m floor?**
$12\times15=180$ tiles.

**8. Paint a 10 m×120 m wall; one gallon covers 35 m².**
Area = 1200 m²; $\frac{1200}{35}$
= 34.29, so buy 35 gallons.

**9. Paint a 10 m×4 m wall; one litre covers 7 m².**
Area = 40 m²; $\frac{40}{7}$
= 5.71, so buy 6 litres.

**10. A trapezoidal window has parallel sides 3 m and 1.5 m and height 2 m.**

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Area $= \frac{3+1.5}{2} \times 2$
$= 4.5\ m^2$.

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